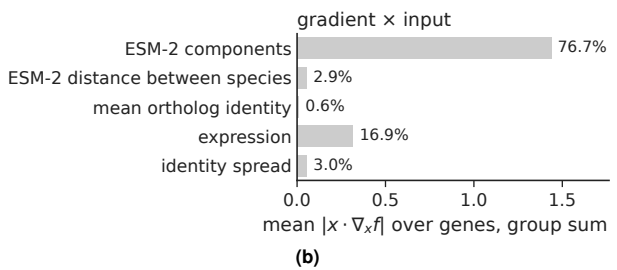
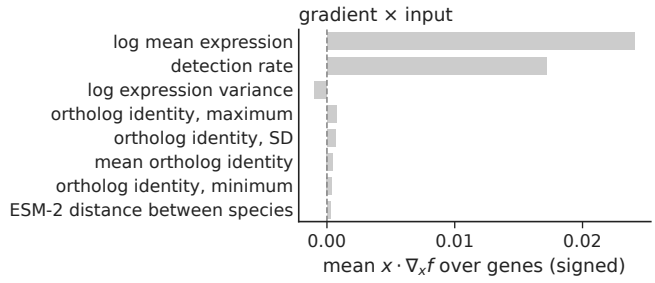

ablation (group set to zero)
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
38.1%
8.6%
1.6%
47.1%
4.5%
0.0
0.1
0.2
0.3
mean |Δ correction| over genes

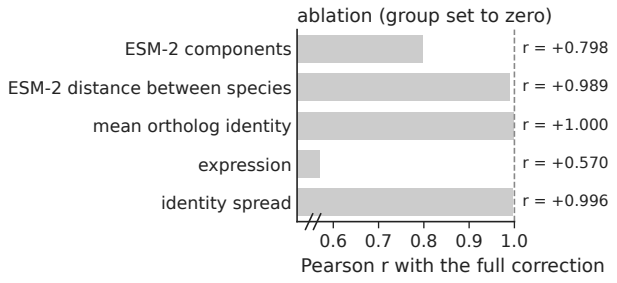
(a)


gradient × input
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
76.7%
2.9%
0.6%
16.9%
3.0%
0.0
0.5
1.0
1.5
mean |x · ∇ₓf| over genes, group sum

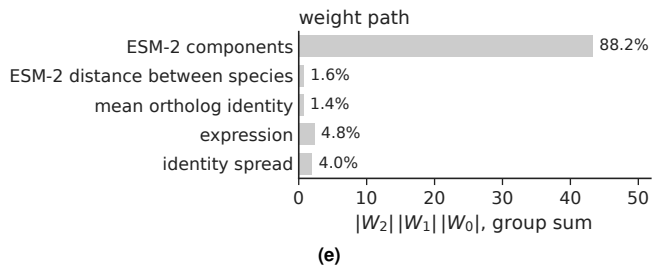
(b)


gradient × input
log mean expression
detection rate
log expression variance
ortholog identity, maximum
ortholog identity, SD
mean ortholog identity
ortholog identity, minimum
ESM-2 distance between species
0.00
0.01
0.02
mean x · ∇ₓf over genes (signed)

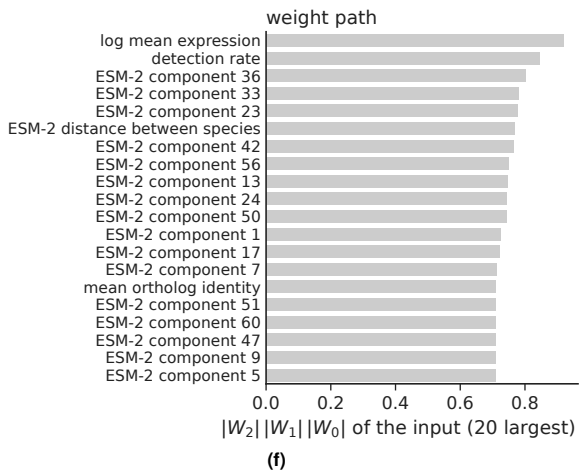
(c)


ablation (group set to zero)
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
r = +0.798
r = +0.989
r = +1.000
r = +0.570
r = +0.996
0.6
0.7
0.8
0.9
1.0
Pearson r with the full correction

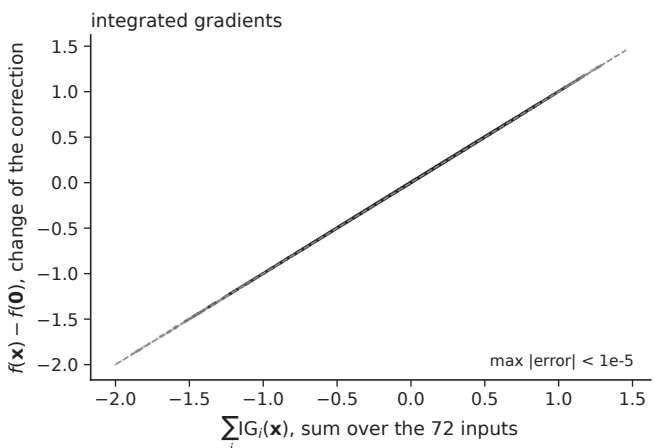
(d)


weight path
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
88.2%
1.6%
1.4%
4.8%
4.0%
0
10
20
30
40
50
|W₂| |W₁| |W₀|, group sum

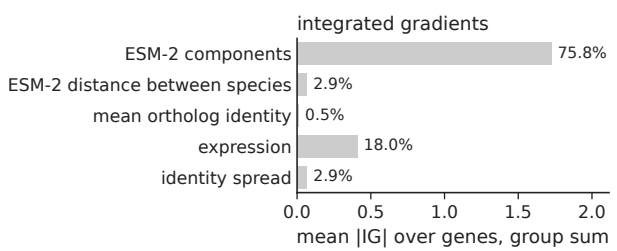
(e)


weight path
log mean expression
detection rate
ESM-2 component 36
ESM-2 component 33
ESM-2 component 23
ESM-2 distance between species
ESM-2 component 42
ESM-2 component 56
ESM-2 component 13
ESM-2 component 24
ESM-2 component 50
ESM-2 component 1
ESM-2 component 17
ESM-2 component 7
mean ortholog identity
ESM-2 component 51
ESM-2 component 60
ESM-2 component 47
ESM-2 component 9
ESM-2 component 5
0.0
0.2
0.4
0.6
0.8
|W₂| |W₁| |W₀| of the input (20 largest)

(f)


integrated gradients
f(x) − f(0), change of the correction
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
max |error| < 1e-5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
∑ᵢ IGᵢ(x), sum over the 72 inputs

(g)


integrated gradients
ESM-2 components
ESM-2 distance between species
mean ortholog identity
expression
identity spread
75.8%
2.9%
0.5%
18.0%
2.9%
0.0
0.5
1.0
1.5
2.0
mean |IG| over genes, group sum

(h)